Problem Set 4 Solutions
ECO3121 - Fall 2023

December 15, 2023

Question 1

A researcher has a panel data set for crop yield of 1000 pieces of land parcels in rural China over year 2000-2010. He considers the following model:

$$Crop_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_{it} + \gamma_t + u_{it}$$

Suppose the researcher has data for $Crop_{it}$ and X_{it}; but he does not have data on Z_{it}:

1. (8 points) Suppose the researcher knows that $Z_{it} = Z_t$ during this period, i.e., Z_{it} is the same for all land parcels but changes over time. Give an example for what Z_t is modeling. Explain how the researcher should estimate with the data set he currently has.

   Answers:A factor that can be modeled as a time fixed effect in this context can be given national agricultural tax policies (macroeconomic condition or business cycle). Clearly, this factor would affect different villages in the same way, and it's likely to change over time. [Open Question to give an example]
   The model is now given by:

   $$Y_{it} = \beta_1 X_{it} + \lambda_t + u_{it}$$

   The fact that $\lambda_t = \beta_2 Z_t + \gamma_t$ is not observable, poses an omitted variable problem in estimating our model by OLS. Let me then define:[Demean Approach, you can also use dummy variable approach or change(first difference) of OLS approach.]

   $$\overline{Y}_t = \frac{1}{N}\sum_{i=1}^{N} Y_{it} = \beta_1 \overline{X}_t + \lambda_t + \overline{u}_t$$

   where X_t and u_t are the analogues of Y_t for X_{it} and u_{it}.
   Let's now consider the demeaned version of the model above:

   $$Y_{it} - \overline{Y}_t = \beta_1 (X_{it} - \overline{X}_t) + u_{it} - \overline{u}_t$$

   Let $\tilde{Y}_{it} = Y_{it} - \overline{Y}_t$ and, in the same fashion, construct $\tilde{X}_{it}$ and $\tilde{u}_{it}$ to rewrite:

   $$\tilde{Y}_{it} = \beta_1 \tilde{X}_{it} + \tilde{u}_{it}$$

   As before, we just need to estimate this model using OLS to get $\hat{\beta}_1$.

2. (8 points) Suppose the researcher knows that $Z_{it} = Z_i$ during this period, i.e., Z_{it} is the same for all years but changes over land parcels. Give an example for what Z_i is modeling. Explain how the researcher should estimate with the data set he currently has.

   Answers: A factor that can be modeled as a village fixed effect in this context can be given by the geographic location of the village itself, for example, rainfall, weather condition, etc.[Open Question to give an example]
   The model is now given by:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \beta_2 Z_i + \gamma_t + u_{it}$$

   or

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \mu_i + \gamma_t + u_{it}$$

   The fact that μ_i and γ_t are not observable, poses an omitted variable problem in estimating our model by OLS.
   Then, the suggested estimation method for two fixed effect is $(n-1)$ and $(T-1)$ binary regressor OLS regression. [You can also use "entity and time-demeaned OLS regression", but should be careful when demeaning for twice to give a complete math induction] Create the binary variables for entity $D2_i, ..., Dn_i$ and for time $B2_i, ..., BT_i$.
   The estimation model is:

$$Y_{it} = \beta_0 + \beta_1 X_{it} + \mu_2 D2_i + ... + \mu_n Dn_i + \gamma_2 B2_i + ... + \gamma_T BT_i + u_{it}$$

Question 2

Following a national poverty alleviation program in Gansu province, in July 2005, all farmers in Gansu province received free fertilizer. Thus, this policy experiment induced a geographical allocation of fertilizers that can be presumed exogenous in a crop yield regression. Let Y_i^0 and Y_i^1 denote the crop in location i before and after the attack. Let D_i be a dummy variable that takes the value 1 if land parcel i is in Gansu province, 0 otherwise. We would like to estimate the treatment effect of the fertilizer usage on crop yield with the difference-in-difference estimator.

1. (6 points) Write down this difference-in-difference estimator as a function of $\{Y_i^0, Y_i^1, D_i\}$ for $i = 1, ..., n$.

   Answers:

   - *The differences-in-difference estimator for the causal effect of poverty alleviation program corresponds to β_1 inthe regression,*

$$\Delta Y_i = \beta_0 + \beta_1 D_i + u_i$$

   for $Y_i = Y_i^1 - Y_i^0$.

- *Alternatively, we can write the treatment effect by* $(\bar{Y}^{treated,1} - \bar{Y}^{treated,0}) - (\bar{Y}^{control,1} - \bar{Y}^{control,0})$*, where* $\bar{Y}^{treated,1}$ *is the sample average of* Y_i^1 *among* $D_i = 1$*, i.e., farmers in Gansu province, and* $\bar{Y}^{control,1}$ *is the sample average of* Y_i^1 *among* $D_i = 0$*, i.e., neighborhoods without a Jewish intuition nearby, and the other two terms are defined similarly.*

2. (6 points) What is the key assumption for this estimator to be an unbiased estimator of the treatment effect?

   Answers: The key assumption is the ***parallel*** *trend assumption: without the program, the crop yield change* $Y_i = Y_i^1 - Y_i^0$ *on average would have been the same for the treatment group and the control group.*

3. (6 points) Suppose after the policy intervention, farmers in Gansu province works harder and thus higher crop yield is expected. Does the difference-in-difference estimator under and over estimate the treatment effect. Explain.

   Answers:

   - *Over-estimate*
   - *The reason being that an increase in the effort should, all else equal, increase crop yield - this being correlated with fertilizer usage,* β_1 *would capture said effect.*
   - *Alternatively, we can explain this through violation of the parallel trend assumption. In this case, the crop yield in the treatment group would have gone up even without the treatment. Attributing this change to the effect of the treatment will result in an over estimation of the effect of the poverty alleviation program.*

Question 3

We are going to replicate a study conducted by Card and Krueger in 1994 that investigate the relationship between a rise in minimum wage and employment.

Economic theories have long suggested that increases in the minimum wage lead to a reduction in the employment for at least two reasons: Businesses are less likely to hire and will rather invest in other resources that are now cheaper because of wage increase. Higher salaries will induce businesses to raise their prices to compensate their greater costs; as prices increase, we expect fewer buyers, which will lead to lower demand and employment.

These theories have found mixed support but the discussion is still very much open within the policy world, as states discuss the opportunity to rise their minimum wage to help local populations to face increasing living costs. Discussions are currently occuring in New Jersey and Illinois to raise the minimum wage to 15$/hour (New york has successfully passed this same raise in 2018).

One of the first study looking at this policy problem was Card and Krueger's. They applied a difference-in-difference the design to look at two groups of fast-food restaurants:

fast-food restaurants in New Jersey where the minimum wage increased from 4.25$ to 5.05$ per hour (treatment group) AND fast-food restaurants in Pennsylvania where the minimum wage did not increase (control group). They collected employment data before and after the minimum wage was approved.

Table 1: Variable Description

Variable name	Description
ID	Unique identifier for fast food
Treatment	Pre-treatment (=0) and post-treatment (=1)
Group	1 if NJ (treatment); 0 if PA (Control)
Empl	# of full time employees
C.Owned	If owned by a company (=1) or not (=0)
Hours.Opening	Number hours open per day
Soda	Price of medium soda, including tax
Fries	price of small fries, including tax
Chain	1 = BK, 2 = KFC, 3 = Roys, 4 = Wendys
SouthJ	South New Jersey
CentralJ	Central New Jersey
NorthJ	North New Jersey
PA1	Northeast suburbs of Philadelphia
PA2	Easton and other PA areas
Shore	New Jersey Shore

1. Explain why comparing New Jersey vs. PA after the change in the minimum wage may not be a good estimation for the treatment effect. (1)

   Answers: It is confounded by the unobserved differences between NJ vs. PA. (Give a specific example or draw a graph to illustrate to get full credit)

2. What is the regression model you'd like to estimate? (1 point)

   Answers: The regression model is specified as follows:

   $$Empl = \beta_0 + \beta_1 Treatment + \beta_2 Group + \beta_3 Treatment * Group + \alpha_1 X + u$$

   where X stands for the series of control variables.

3. What is the key assumption of the estimation method you use? (1 point)

   Answers: Key assumption: parallel trend assumption, i.e. without the treatment, the treated group and the control group were to move in the parallel trend over time.

4. The data is in the blackboard (bb) Assignment 4 file folder titled *DID_Example.csv*, and the variables are defined as in Table 1. Run the regression model you proposed in 2. and report the regression results. (Please include control variables that you think necessary) (2 point)

Answers: Stata command:
reg Empl i.Treatment i.Group i.Treatment # i.Group X
(X denotes the series of control variables)
or
*gen Treatment_Group=Treatment * Group*
reg Empl Treatment Group TreatmentGroup X

5. Explain why this is an unbiased estimation of the treatment effect. (Hint: You can use a graph if necessary) (1 point)

   *Answers: Draw the graph and the 2*2 table as in the slides, and show that the interaction term estimate the "true" treatment effect compared.*